

Solid Edge to Teamcenter Import Data Preparation Utilities

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Preface

This manual describes the tools and processes used to prepare unmanaged Solid Edge documents for import into the Teamcenter environment using the Add to Teamcenter Data Preparation Utilities delivered on the Solid Edge V20 software installation media, and installed with Solid Edge.



This document supplements the Solid Edge Embedded Client User's Guide for Administrators and Clients (*SEECUsersGuide.pdf*) and the Solid Edge online help installed with Solid Edge.

Audience

This manual is intended for Solid Edge Embedded Client administrators in an unmanaged Solid Edge environment who want to import the unmanaged files into a Teamcenter Engineering or Teamcenter Express managed environment. An understanding of Solid Edge and Teamcenter Engineering concepts are required.



If you are importing unmanaged files into a prepopulated Teamcenter database, please contact your UGS representative for consultation.

Conventions

The following icons represent tips, notes, caution, and warning messages used within this manual:



A tip icon indicates information that helps you apply the techniques and procedures described in the text.



A note icon identifies general instructions or comments that need to be emphasized.



A caution icon identifies practices that can either produce results contrary to what you expect or result in damage to software or data.



A warning icon identifies practices that could result in permanent loss of data or software.

Teamcenter Engineering Documentation

Teamcenter documentation is provided outside the context of this document as online help and as printable manuals:

- You can access online help by choosing Help from the menu bar of a Teamcenter rich client application or by clicking one of the links under the Help icon in the Teamcenter Engineering thin client user interface.

- You can access the printable manuals from the Teamcenter Engineering Documentation CD-ROM. To view the PDF-formatted manuals, use Adobe Acrobat Reader.

Submitting Comments

Please feel free to give us your opinion of the usability of this manual, to suggest specific improvements, and to report errors. Mail your comments to:

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To submit your comments online, you can also use the UGS GTAC online support tools at <http://support.ugs.com>.

Chapter

1 *Introduction*

Importing thousands of unmanaged Solid Edge documents into Teamcenter can be a time-consuming process. Careful attention is needed to analyze existing documents, modify file properties based on your desired managed structure, and correct broken links and duplicate filenames discovered during analysis. The Solid Edge to Teamcenter Data Preparation Utilities can assist you in this process and prepare your data for importing into a managed environment using the Add to Teamcenter application.



The Add to Teamcenter application is the only mechanism to import unmanaged Solid Edge documents into a Teamcenter-managed environment.

This document provides you with information about how to:

- Obtain and access the *Solid Edge to Teamcenter Data Preparation Utilities*.
- Understand the relationship between Solid Edge file properties and Teamcenter attributes, which is necessary for analyzing your existing data.
- Perform an analysis of your existing data using the Analyze Files utility.
- Fine-tune the data generated by the analysis through the use of the analysis spreadsheet, so that it correctly represents the data that will be imported into Teamcenter.
- Update the unmanaged files with changes identified by the Modify Files utility.
- Repair broken links using the Link Fixup utility.
- Create a spreadsheet of files to import into Teamcenter.
- Process the data in the spreadsheet in a trial-run environment.
- Import the modified data from Solid Edge into Teamcenter using the Add to Teamcenter utility.

Chapter

2 *Getting Started with the Solid Edge to Teamcenter Import Data Preparation Utilities*

The *Solid Edge to Teamcenter Data Preparation Utilities* are delivered on the Solid Edge V20 software installation media, and they are installed with Solid Edge. You do not need to install them separately.

A Few Considerations Before Running the Utilities

The processing of unmanaged documents by the *Solid Edge to Teamcenter Data Preparation Utilities* takes place on the client system running the utilities. Considerations prior to running the utilities are:

- A minimum of 4gb of memory is recommended on the client system.
- Available disk space 1.5 times the size of the dataset being processed is recommended for the client system.
- Microsoft Office Professional 2003 or 2007 must be installed on the client to run the *Teamcenter Data Preparation Utilities*.



Due to a limitation in Excel, the maximum number of files that can be processed at one time by the utilities is 65,536.

- The *Solid Edge to Teamcenter Data Preparation Utilities* are not compatible with the Windows 2000 operating system.
- Identifying a target dataset for importing, such as a specific folder of data, is recommended over a mass importation of data into a Teamcenter-managed environment.



If you are importing more than 20,000 documents, it is recommended that you contact your reseller or UGS representative to discuss best practices.

- Identifying the non-Solid Edge files you want to import before running the utilities is recommended. If the non-Solid Edge files are not linked to Solid Edge files, they should be imported into Teamcenter outside the Add to Teamcenter environment. Consult Teamcenter documentation for importing the non-linked, non-Solid Edge files into Teamcenter.

- The computer containing data being analyzed or imported should not be accessible by those actively modifying the documents in a production environment. The Add to Teamcenter utility has the functionality necessary to recognize files that have been changed in production during the import time-frame and then add the delta to the database.
- A unique Teamcenter user id should be created for running Add to Teamcenter. File ownership inside the Teamcenter database is determined by this user id.

Understanding Teamcenter Terms

Teamcenter Objects

Teamcenter uses objects to store information that describes each Solid Edge document, in addition to storing the document or file itself. An *Item* is a commonly used Workspace object that captures the metadata describing all the associated revisions. An *Item Revision* object captures the revision-specific metadata. The document is captured in a *Dataset* object and connected to the Item Revision.

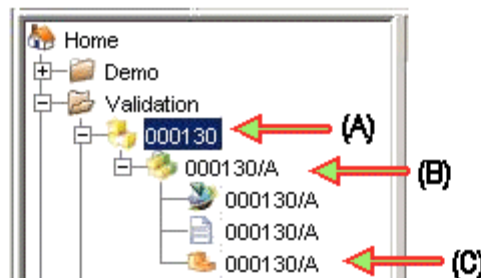
It is metadata that is useful in data retrieval and reuse. The Item, Item Revision and Dataset combine to fully describe the associated document.



Datasets have the ability to capture versions of a document.

If you want to think of these objects in terms of hierarchy, from top to bottom, it would be Item, Item Revision, Dataset. You can view this hierarchy from the Teamcenter Client. Each object is represented by a unique image, and as you expand each level, you can see the structure. Once you reach the Item Revision, you may see more than one Dataset. When your business process requires the Solid Edge draft file to have the same Document Number as its 3D part (.par), assembly (.asm), sheet metal (.psm) or weldment (.pwd) file, you will have a draft (.dft) file with the corresponding number under the Item Revision. This practice impacts how a company does Revisions.

Each object must have a unique identifier (ID). For Items, this is called an Item ID. In the following example, the Item ID for the Item (A) is 000130. The Item Revision (B) for this item is /A, which adds a unique Revision designation, followed by the Dataset (C) and its Name.



The attributes that describe these objects are often the same, but when applied to the Item, Item Revision or Dataset, they provide uniqueness. For example, since the Name attribute is used by these three objects, to get a specific piece of information you can specify the Item Name, Item Revision Name, and Dataset Name.

For more information on Teamcenter terminology, see the *Getting Started With Teamcenter Express Guide* (pub number tcx00002) or the *Teamcenter Express Installation and Configuration Guide* (pub number tcx00001).

Understanding the relationship between Solid Edge Properties and Teamcenter Attributes

Attribute mapping defines what Solid Edge document properties you exchange with Teamcenter. In Teamcenter, the primary attributes used to track documents are *Item ID*, *Revision ID*, and *Item Name*.

The Solid Edge properties that are automatically mapped to Teamcenter attributes are displayed in Solid Edge on the Project tab of the Properties dialog box. You can access the Properties dialog box by selecting File→File Properties and clicking the Project tab in Solid Edge.

Solid Edge property	Teamcenter attribute	Maximum character length
Document Number	Item ID	32 characters
Revision Number	Revision ID	32 characters
Project Name	Item Name	32 characters

By utilizing the Solid Edge to Teamcenter Import Data Preparation Utilities to prepare your data to properly map existing Solid Edge properties to these attributes before adding them to Teamcenter, you can minimize future efforts.

If any of the Solid Edge properties contain more than 32 characters, the property value will be truncated to the maximum character length upon import in to Teamcenter. Additionally, if the Document Number plus the Revision contain a total of more than 32 characters you will be notified of an error on import into Teamcenter. The error is recorded in the log file generated by the import utility in the *\Application Data\Unigraphics Solutions\Solid Edge\Version 20\Log Files* folder.

When you import the unmanaged files into the managed environment, blank Solid Edge Document Number properties, are automatically assigned an Item ID by Teamcenter. It is recommended that you populate the Document Number, Revision Number, and Project Name fields prior to loading your data into Teamcenter. If these fields are properly populated, your Teamcenter database will be more accurately populated from the start.



When unmanaged 3D documents using the same Document Number property are imported into Teamcenter, a unique document number is assigned to each 3D file being imported. As a result, in Teamcenter there can only be one non-Draft Solid Edge file within each item revision. However, multiple Draft files using the same Document Number property can accompany a single 3D file in an item revision.

You can create custom Solid Edge properties to map to Teamcenter attributes such as Item Type and Dataset Description. Refer to the Solid Edge Help topic, *Create Custom Document Properties* for specific instructions.

Solid Edge property	Teamcenter attribute	Maximum character length
Teamcenter Item Type	Item Type	32 characters
TC Engineering Description	Dataset Description	240 characters



The Solid Edge property to Teamcenter attribute mapping is case sensitive. When you create custom properties, be sure to use the capitalization shown in the table.

The Data Preparation utility, Analyze Files, provides an automated approach for mapping the Solid Edge properties to Teamcenter attributes.

Accessing the Solid Edge to Teamcenter Data Preparation Utilities

You can access the *Solid Edge to Teamcenter Data Preparation Utilities* directly from the folder system on your computer. You do not need to install them separately.

Step 1: Browse to the `\Program Files\Solid Edge V20\Custom\tc data prep utils` folder.

Step 2: Double-click the sub-folder that contains the utility you want to run.

- [Data Prep Utilities - Analyze Files](#)

- Performs a filename length analysis.
- Evaluates Solid Edge files for duplicate document number values, duplicate filenames, empty revision entries, empty project names, and broken links.
- Evaluates the Solid Edge to Teamcenter property mapping.



If you use custom Solid Edge data properties, define the custom properties for use in Teamcenter using row 7 of the spreadsheet *DataPrepUtilitiesTemplate.xls* located in this folder.

- [Data Prep Utilities - Modify Files](#)

Updates unmanaged Solid Edge files based on entries in the spreadsheet that is produced as the output of file analysis. New property values are saved in the Solid Edge files.

- [Data Prep Utilities - Link FixUp](#)

Repairs broken links in Solid Edge files before you import them into Teamcenter.

Step 3: Double-click the application executable file in the folder you selected.

The application executable names are:

- *AnalyzeAndBuildSpreadsheet.exe*
- *ModifyIndividualFilesInSpreadsheet.exe*
- *LinkFixUpSearchExcelForReplacement.exe*

Removing previously installed import utilities

If you have a previous version of the Solid Edge to Teamcenter import utilities, use Add Or Remove Programs in the Control Panel to remove **SolidEdgeIntoTeamcenterUtilities** from your computer.

To remove this version of the utilities from your computer, look for **Solid Edge Data Prep Utilities** in the Add Or Remove Programs dialog box.

Chapter

3 *Analyze Solid Edge Files Utility*

A member of the Solid Edge to Teamcenter Data Prep Utilities, the Analyze Files utility evaluates Solid Edge files for duplicate document number values, duplicate filenames, empty revision entries, empty project names, and broken links. You can display the results on your computer, or save the results to a text file.

Some of the features of the utility include:

- Filename length analysis.
- Solid Edge to Teamcenter property mapping evaluation.
- Broken link analysis.
- Full file preparation and test capability.

Workflow overview

The steps required to perform an analysis consist of:

1. Defining any Solid Edge custom properties in the *DataPrepUtilitiesTemplate.xls* spreadsheet.
2. Defining the location of the files for analysis.
3. Defining the criteria for the analysis.
4. Defining the location for the report file.
5. Processing the input.
6. Displaying the results.

Performing a Solid Edge file analysis

Analyze Solid Edge files

Step 1: (Optional) Define your Solid Edge custom properties in row 7 of the *DataPrepUtilitiesTemplate.xls* spreadsheet prior to running the Analyze Files utility.

- ☐ Access the spreadsheet by opening the file *DataPrepUtilitiesTemplate.xls*. This file is located in the *\Program Files\Solid Edge V20\Custom\tc data prep utils\Data Prep Utilities - Analyze Files* folder.
- ☐ On row 7 of the spreadsheet, to the right of the predefined property information, enter your custom property names not already defined in the section displayed in orange. Example: TC Engineering Description.
- ☐ Save and exit the *DataPrepUtilitiesTemplate.xls* spreadsheet.

Step 2: Run the *AnalyzeAndBuildSpreadsheet.exe* application.

The default location for the application is the *\Program Files\Solid Edge V20\Custom\tc data prep utils\Data Prep Utilities - Analyze Files* folder.

Step 3: In the Solid Edge to Teamcenter Data Prep Utilities - Analyze Files dialog box, specify whether the files to be analyzed are located in a spreadsheet or system folder by selecting one of these options:

- Specify Spreadsheet Containing List Of Files To Process
You must identify the location of a spreadsheet containing the list of files being analyzed. The spreadsheet must contain the full path of the location of the file along with the filename in a complete string in the first column starting with the first row.
 - Specify Top Level Folder Containing Solid Edge Files
You must identify the location of the system folder containing the files for analysis.
-  All Solid Edge file types are recognized for analysis using either of the location options.

Step 4: Use the Browse button associated with your selection to locate the spreadsheet or folder specified in the previous step.

Step 5: Verify that the Create Spreadsheet Containing Complete List Of Files check box is selected.

Step 6: (Optional) Refine the analysis by selecting additional, appropriate check boxes.

You can choose from these options to define the criteria for your analysis:

- Crawl Links To Build Complete List Of Solid Edge Files To Be Imported.

- Perform Broken Links Analysis.
- Add Draft File If Found In Same Folder With Same Name.

Step 7: In the Report File box, define a location for the report file generated as output of the analysis.



The report file automatically generated is *AnalysisOfSolidEdgefiles.txt*. By default, the location for the report file is the same location that you defined for the spreadsheet or folder in Step 4.

Step 8: For large data sets, select the **Do Not Sort The Excel Spreadsheet** check box.

Selecting this option will save time and resources necessary to sort the output generated by the analysis.

Step 9: Click OK to start the analysis.

The length of time required for processing depends on the number of files analyzed. When processing is finished, the Report section of the Analyze Files dialog box updates to display the statistics of the files analyzed. Also, an Excel document is displayed. The top portion of the document is the [Analysis Template](#). The bottom portion is the [Analysis Spreadsheet](#).



You may need to expand the Excel document to see both the top and bottom sections.

Step 10: (Optional) To display the report in a separate text file, in the Analyze Files dialog box, in the Select Report Files To Use area, click one of the reporting option buttons.

In addition to displaying the results of the analysis in the dialog box, the results are saved to disk in the location you define in the Report File portion of the dialog box. You can click **View Report Text File** to view the contents of the file on disk.

Step 11: Save the file analysis output to a different name than the default *DataPrepUtilitiesTemplate.xls* and close the Excel document.



If you reopen the spreadsheet, macros may be disabled depending on your Microsoft Office security settings. Discuss your security policy with your system administrator before making changes to your security settings.

Step 12: In the Analyze Files dialog box, click Cancel to close the utility.

This completes the analysis portion of the workflow for using the Solid Edge to Teamcenter Import Data Preparation Utilities.

To fix problems identified in the analysis, use the tools in the [Analysis Template](#) (the top portion) to apply corrections to the files listed in the [Analysis Spreadsheet](#) (the bottom portion).

An example of the types of corrections you might need to apply is renaming existing unmanaged files. The process for [Renaming Files Using the Solid Edge to](#)

[Teamcenter Data Preparation Utilities](#) is described in the following section of this document.

Once any necessary corrections have been made to the unmanaged files, you are ready to repair broken links and update the unmanaged Solid Edge files:

- If broken links were identified in the file analysis, use the Link Fixup utility to repair them. This utility modifies the files directly. To learn how to do this, see [Fixing Links for Solid Edge Files](#).
- To modify the unmanaged Solid Edge files with the changes you specified in the Excel document, run the [Modify Solid Edge Files utility](#).

Renaming files using the Solid Edge to Teamcenter Data Preparation Utilities

Following file analysis, you may find that existing unmanaged files need to be renamed, either to separate the filename and revision number (as is the case with files from Insight), or because of other circumstances. The following procedure describes the steps for using the data preparation utilities to rename unmanaged files before importing them into a Teamcenter-managed environment.

Step 1: Perform a file analysis of unmanaged Solid Edge files using the option to Specify Top Level Folder Containing Solid Edge Files.

When the analysis completes, the Data Prep Utilities Template displays the results. The spreadsheet portion of the template contains rows and columns of cells for correcting or confirming the data prior to importing it into Teamcenter.

Step 2: Specify the new filename and extension in the New Filename column of the analysis spreadsheet.



If your existing filename contains both a document number and a revision, which is often the case when the data being renamed is from an Insight-managed environment, use the Split Filename into Doc. Number and Revision Property command and define a delimiter for separating the two values. Then you can use the Copy Doc Number to New Filename command to create the new filename.

Step 3: In the New Path column of the analysis spreadsheet, specify the location for the new files created during this process.



You can copy and paste the location from the Path column on the spreadsheet.

Step 4: Define the Teamcenter Item Type for each file.

The Item Type, Item, is case-sensitive.

Step 5: Remove all dashes from the spreadsheet cells using the Replace All Instances of the Following String with an Empty String command. Specify the dash (-) as the character to replace in the cell to the right of the command.



This function affects the entire spreadsheet, not just the active cell, row, or column of the spreadsheet.

Step 6: Close and save the spreadsheet. You are prompted to save the spreadsheet to a new name on the Save As dialog box. Save the spreadsheet to the same location as your dataset.

Step 7: In the `\Custom\tc data prep utils\Data Prep utilities - Modify Files` folder, run *ModifyIndividualFilesInSpreadsheet.exe*.

Step 8: Browse to the location of the spreadsheet you defined in Step 6 and click Process.

The utility copies the original files to the new filenames.



Processing time depends upon the number of files identified in the spreadsheet.

Step 9: Once processing is complete, reopen the analysis spreadsheet.

Step 10: Mark the original files for deletion by entering an X in the Delete column of each file copied to a new filename.

Step 11: Select **Delete Documents marked by X**.



This action is performed immediately on the unmanaged files on disk.

☐ Dismiss the notification that files have been deleted.

Step 12: Exit the spreadsheet.

You do not need to save the changes to the spreadsheet.

Step 13: Run the *\Custom\tc data prep utils\Data Prep utilities - Link Fix-Up\LinkFixUpSearchExcelFor Replacement.exe* utility to repair the broken links created when you renamed the files.

☐ On the Link Fix-Up dialog box, browse for the spreadsheet you defined in Step 6.

☐ Locate the folder containing the SE files.

☐ In the Doc Types section of the Link Fix-Up dialog box, select the Parts, Assemblies, and Drawings check boxes.

☐ Define the scope by selecting the All Files in Selected Folder and Sub-folders option.

☐ Click OK to run the utility.

The length of time required for processing depends on the number of files analyzed.

Step 14: When the Link Fix-Up has finished processing, click Cancel to dismiss the Link Fix-Up dialog box.

The documents have been renamed and the links re-established. You are ready to import the files into Teamcenter using the Add to Teamcenter utility described in Chapter 6 – Adding Documents to a Teamcenter Managed Environment.

Analyze Files dialog box

Use the Solid Edge to Teamcenter Data Prep Utilities - Analyze Files dialog box to define the parameters related to the evaluation and modification of your Solid Edge files.

The analysis can produce a spreadsheet-like interface for you to review the files prior to importing them into Teamcenter. It consists of tools to manipulate the data evaluated ([Analysis Template](#), the top portion of the interface) and the results of file evaluation ([Analysis Spreadsheet](#), the bottom portion of the interface).

You can use the Analyze Files dialog box to:

- Define the location of the files to process.
- Determine the scope of the analysis.
- Define the location for the report file generated from the analysis.

Specify Options To Locate File To Process

Defines the location of the data that is evaluated.

Specify Spreadsheet Containing List Of Files To Process

Identifies a list of files for evaluation. The list is in the form of a spreadsheet containing the filenames.

Specify Top Level Folder Containing Solid Edge Files

Defines a top level folder containing the files to be analyzed. All Solid Edge file types are recognized.

Create Spreadsheet Containing Complete List Of Files

Creates an Excel spreadsheet consisting of tools to manipulate the data evaluated (Analysis Template) and the results of file evaluation in spreadsheet form (Analysis Spreadsheet).

Crawl Links To Build Complete List Of Solid Edge Files To Be Imported

When selected, analysis includes a recursive search for external files referenced by Solid Edge files. The recursive search may be outside the folder structure you specified. When deselected, only the folder you specified will be analyzed.



This command is especially useful when you want to find links to files like Standard Parts held on servers.

Perform Broken Links Analysis

Selecting the check box enables the analysis for broken links.

Add Draft File If Found In Same Folder With Same Name

During file analysis the draft file is added if it is found with the same name as the part file.

Report File

Defines the location for the text file containing the results of the analysis. The default filename of the report file is *AnalysisOfSolidEdgefiles.txt*

Reports

Displays a summary of the results of the file analysis.

Select Report Log Files To View

Controls the display method for the listed documents.

- View Report Text File
- View Simple Broken Links Report
- View Detailed Broken Links Report

Do Not Sort The Excel Spreadsheet

When selected, the results are not sorted. This is useful with large data sets because Excel does not spend time or resources sorting the results.

OK

Starts analysis processing. Status messages display at the bottom of the Solid Edge to Teamcenter Data Prep Utilities - Analyze Files dialog box.

Cancel

Closes the Solid Edge to Teamcenter Data Prep Utilities - Analyze Files dialog box.

Analysis Template

The Analysis Template displays the results of data analysis in spreadsheet form and contains tools for manipulating data in preparation for importing into Teamcenter.



The default location of the Analysis Template file *DataPrepUtilitiesTemplate.xls* is the *\Program Files\Solid Edge V20\Custom\tc data prep utils\Data Prep Utilities - Analyze Files* folder.

You can use the Analysis Template to:

- Identify documents to delete from disk.
- Prepare files with property data prior to importing into Teamcenter.
- Perform trial runs on data importation prior to loading the data into Teamcenter.

Color Key

This portion of the dialog defines the color coding associated with the output of the data analysis. The color coding makes it easier to identify potential problem areas that require attention.

Red Denotes files found with duplicate filenames.

Yellow Denotes files found that contain duplicate document number property values.

Delete Documents Marked By X

Scans the Delete column of the spreadsheet for those marked with an X. When processed, these files are immediately deleted from disk.



Deleting files from disk can create broken links. Use care when marking files for deletion.

Copy Filename (w/o ext.) To Only Empty Doc. Number

Copies the existing filename of a document without its extension to the Document Number property. As a result, once the data is imported into Teamcenter, the filename is used for the document number.

Copy Filename (w/o ext.) To Project Name

Copies the existing filename of a document without its extension to the Project Name property. As a result, once the data is imported into Teamcenter, the filename is used for the project name.

Compute Longest Character For Each Column

Calculates the total character length of each field and reports it for evaluation.



There is no logic applied to compare the values reported to the maximum character limit of fields within Teamcenter. You must manually verify the maximum character limit.

Split Filename Into Doc. Number And Revision Property

Looks for a delimiter defined in the delimiter column of the spreadsheet and divides the filename into two portions. Data to the left of the delimiter is used in the Document Number property definition. Data to the right of the delimiter is used in the Revision property definition.

Copy Doc. Number to New Filename

Copies existing document numbers to new filenames. Existing data is not modified.

Remove Colored Cells

Removes the color coding of cells in the spreadsheet. This command should be performed prior to Rechecking for Duplicates in Document Number and Filename.

Truncate Maximum Number of Characters—Max # —Applied to Column

Reduces the character length of data found in the column you specify to the number of characters you specify. Use the Column field to define the numeric column location for this action.

Recheck For Duplicates In Document Number And Filename

Evaluates modified data to ensure the property information is correct and contains no duplicate data prior to importing into Teamcenter.

AutoIncrement Document Number—Prefix—Start With

Increments the existing document number with the prefix and starting value you specify.

Append String To Existing Document Number—Before—After

Appends the text string you define before and after the existing document number.

Parse Filename By Start/End String Location—Starting—To—Place Result In Column

Sorts the data by the value you supply.



The value for the destination column is the numerical value as counted left to right on the spreadsheet.

**Find Files With This String Somewhere In The Filename-
Place the Following String in Corresponding Column**

Finds files with the string you type in the box to the right, and then places another string you specify in the column you specify.



Standard parts usually have something unique in the filename. This function can be used to add the word “Standard” in the Teamcenter Item Type column for that file.

Replace All Instances of the Following String with an Empty String

Replaces the contents of cells containing the identified string with an empty string so the value is null.



The null value is identified with a dash (-). You should replace the dash with valid information prior to importing the data into Teamcenter.

Populate Draft Document Numbers Based On Linked 3D File

Looks for the draft file and finds the linked part file to populate with the document number and revision.

Populate 3D Files From Draft Files Based On Having Same Filename

Looks for the part file and finds the linked draft to populate.

Make Listed Files Read-only

Immediately sets all listed unmanaged files to read-only status.

Make Listed Files Available

Immediately sets all listed unmanaged files so that they are not write-protected.

Remove Import Time Stamp

Immediately deletes the timestamp from each listed file to allow the same data to be imported more than once into the Teamcenter database.

Reset Status To Available

Immediately resets the Solid Edge status property of selected files to available.

Search For Drafts Linked To Multiple 3D Files

Identifies all draft files that contain multiple links to Solid Edge 3D files. When found, it places the text, “Linked to Multiple 3D,” in column 5 of the row where the draft file is listed. You manually define which document number is used.

Analysis Spreadsheet

The Analysis Spreadsheet displays the results of data analysis in spreadsheet form in the bottom portion of the Analysis Template. The Analysis Spreadsheet contains columns of cells for correcting or confirming data prior to importing into Teamcenter.

You can use the Analysis Spreadsheet to:

- Define custom properties prior to running the analyze tool.
- Mark documents to delete from disk.
- Define property data prior to importing into Teamcenter.

Path

Displays the full path of the folder containing the analyzed file.

Filename

Displays the existing filename of the analyzed file.

New Path (if necessary)

Defines a new folder location for the processed file.

New Filename

Defines a new filename for use in conjunction with copy.

Delete

Marks the file for deletion using the X in conjunction with the **Delete Documents Marked By X** option.

Document Number

Displays the document number that will be imported in to Teamcenter.



Do not modify the column header for this column.

Revision

Displays the revision that will be imported into Teamcenter.



Do not modify the column header for this column.

Project Name

Displays the project name that will be imported into Teamcenter.



Do not modify the column header for this column.

Teamcenter Item Type

Displays the Teamcenter Item Type property. The Teamcenter Item Type string is localized and it is case-sensitive.



Only modify the column header for this column if localizing outside of English. This is a custom property and is case-sensitive.

Remaining Custom Properties

Defines the custom properties to be pulled from Solid Edge in the header of row seven of the spreadsheet. This string is case-sensitive.



Define any custom properties prior to running the Analyze tool. You can access this spreadsheet by opening the file *DataPrepUtilitiesTemplate.xls*. This file is located in the folder where you loaded the tools as specified in the setup wizard.

Chapter

4 *Modify Solid Edge Files Utility*

A member of the Solid Edge to Teamcenter Data Prep Utilities, the Modify Files utility updates Solid Edge files based on entries in the Analysis Spreadsheet. New property values are saved in the unmanaged Solid Edge files.

Workflow overview

The steps required to modify the data using the Analysis Spreadsheet consist of the following:

1. Defining the location of the Analysis Spreadsheet to be used for data manipulation.
2. Processing the input.
3. Displaying the results.

Modifying Solid Edge files

The Modify Files utility modifies the files listed in the Analysis Spreadsheet (bottom portion) of the Excel document according to the touch-up specifications listed in the Analysis Template (top portion) of the Excel document.

Step 1: Double-click the *ModifyIndividualFilesInSpreadsheet.exe* application.

The default location for the application is the *\Program Files\Solid Edge V20\Custom\tc data prep utils\Data Prep Utilities - Modify Files* folder.

Step 2: In the Solid Edge to Teamcenter Data Prep Utilities - Modify Files dialog box, click Browse.

Step 3: Locate the Excel spreadsheet on your computer that contains the unmanaged Solid Edge files to be modified.

This file was generated by the [Analyze Files utility](#). The default location for the spreadsheet is the *\Program Files\Solid Edge V20\Custom\tc data prep utils\Data Prep Utilities - Analyze Files* folder. The default name for the spreadsheet is *DataPrepUtilitiesTemplate.xls*.

Step 4: Click Process.

The length of time required for processing depends on the number of files to process. A status notification is displayed to let you know when processing is complete, and the graphic window updates to display the results.

The results of processing the data are also written to a log file.

Step 5: (Optional) Click View Logfile to view the contents of the log generated during modification of the spreadsheet data.

Modify Files dialog box

The Solid Edge to Teamcenter Data Prep Utilities - Modify Files dialog box changes the files as defined in the Analysis Spreadsheet.

You can use the Modify Files dialog box to:

- Define the location of the Analysis Spreadsheet containing the data to be modified.
- Process the files saving new property values in the Solid Edge files as defined in the Analysis Spreadsheet.
- View the results of the process in the log file.

Browse

Defines the location of the Excel spreadsheet used for modifying the data to be imported into Teamcenter.

Report Window Pane

Displays the results of the modification of data.

View Logfile

Displays the contents of the log file generated during the processing of the data.

Chapter

5 *Link FixUp Utility*

A member of the Solid Edge to Teamcenter Data Prep Utilities, the Link FixUp utility repairs links in Solid Edge files before you import them into Teamcenter. The Link FixUp utility locates the Excel spreadsheet produced from the data analysis process and defines the parameters used for repairing broken links.

Workflow overview

The steps required to use the Link Fix-Up utility to correct broken links consist of:

1. Defining the document types to be processed.
2. Defining the scope of files to be processed.
3. Displaying the log file created during file processing.

Fixing links in Solid Edge files

Step 1: Double-click the *LinkFixUpSearchExcelForReplacement.exe* application.
The default location for the application is the *\Program Files\Solid Edge V20\Custom\tc data prep utils\Data Prep Utilities - Link FixUp* folder.

Step 2: In the Solid Edge to Teamcenter Data Prep Utilities - Link FixUp dialog box, click Browse For Spreadsheet.

Step 3: Locate the Excel spreadsheet on your computer that lists the Solid Edge files to be repaired and the specifications for updating them.

This file was generated by the Analyze Files utility. The default location for the spreadsheet is the *\Program Files\Solid Edge V20\Custom\tc data prep utils\Data Prep Utilities - Analyze Files* folder.



For each broken link, the spreadsheet must define both the old file location and the old link as well as the new file location and the new link. Columns 3 and 4 are used for the new path and new link definition.

Step 4: In the Solid Edge to Teamcenter Data Prep Utilities - Link FixUp dialog box, click Locate Folder Containing SE Files.

Step 5: Browse to the folder that contains the Solid Edge files to be repaired and click OK.

- Step 6:** In the Doc Types area of the Link FixUp dialog box, select one or more document types that you want to modify.
- Step 7:** in the Define Scope area of the Link FixUp dialog box, define the scope of the files to be processed by setting one of the options.
- Step 8:** Click OK to modify the files.
- Step 9:** (Optional) Click View Logfile to display the information generated during the Link Fixup process in a separate window.

Link FixUp dialog box

The Solid Edge to Teamcenter Data Prep Utilities - Link FixUp dialog box identifies and corrects broken links noted during file analysis.

Use the Link FixUp dialog box to:

- Define the location of the Analysis Spreadsheet.
- Define the location of the Solid Edge files containing broken links.
- Determine the file types to be affected by the link repair.
- Define the scope of files to be affected by the link repair.

Browse For Spreadsheet

Locates the Excel spreadsheet defining the corrected links.



For each broken link, the spreadsheet must define both the old file location and the old link as well as the new file location and the new link. Columns 3 and 4 are used for the new path and new link definition.

Locate Folder Containing SE Files

Defines the folder containing the files to be processed with link corrections.

Doc Types

Filters the types of documents to be processed.

- Parts—Restricts documents to part files (.par).
- Assemblies—Restricts documents to assemblies (.asm).
- Drawings—Restricts document to draft files (.dft).

Define Scope

Sets the parameters of the files to be processed.

- All Files In Selected Folder ONLY—Restricts the files to be processed to only the files in the selected folder.
- Selected Files ONLY—Restricts the files to be processed to only those selected.

- **All Files In Selected Folder And Subfolders**—Processes all files in the folder selected and its subfolders.

View Logfile

Displays the contents of the log file created during the link correction process.

Chapter

6 *Adding documents to a Teamcenter managed environment*

You can add multiple documents and folders to the Teamcenter environment using the Add To Teamcenter utility.

Using Add To Teamcenter

You can import multiple unmanaged Solid Edge files into a Teamcenter database without starting the Solid Edge application using the *AddToTeamcenter.exe* utility. Before running the executable, ensure that you are able to successfully create and save Solid Edge files. Then prepare your unmanaged files for the managed environment using the tips found in Preparing to Add Documents to Teamcenter.

Running Add To Teamcenter

The Add To Teamcenter utility, *AddToTeamcenter.exe*, is delivered to the Solid Edge V20\Program folder. To run the executable, double click *AddToTeamcenter.exe*. When you run the executable, the [Add to Teamcenter dialog box](#) is displayed providing you options for loading your files into the Teamcenter database.

Overwriting Documents Using Add To Teamcenter

The *AddToTeamcenter.exe* utility compares the item number, revision, and dataset values of a document with those existing in the Teamcenter database. As a result, you can import files over files that already exist in the Teamcenter database. Add To Teamcenter adds the import time to the original file, and on subsequent loads, it checks to see if the modified date is after the import date. If the document being imported has not been modified since the last import, it is recognized as already being in the database and is not re-added to Teamcenter on subsequent imports. However, if the unmanaged file is modified after its initial import into the database, on subsequent imports you can choose one of these options:

- Check out the document and overwrite it.
- Check out all files that exist and overwrite them.
- Leave the file and use the file already in the database.
- Leave all files and use the ones that already exist in the database.
- Cancel the import process.

In the event the Add To Teamcenter comparison discovers an item number that already exists in the database, it compares the file being imported to all revisions, datasets, and files that comprise the dataset. If a match is found, you are given the opportunity to determine the action on the file being imported. If no match is found, the file is assigned a new item number and a record of the action will be placed in the log file.

For example, assume the following information is in Teamcenter:

```
Item ID 000100
Revision A
Dataset SE Part 000100/A
Named Reference part1_04112007.par
Revision B
Dataset SE part 000100/B
Named Reference 000100.par
```

When you import a file with 000100 as the document number, the name of the file being imported is compared to all named references (in this example, part1_04112007.par and 000100.par). If it matches, you are given the option to overwrite the file. If you choose not to overwrite, the file is assigned a new item ID, which is recorded in the log file.

Resuming Add To Teamcenter

When using the *AddToTeamcenter.exe* utility, some requested actions may need further attention and an Add To Teamcenter notification dialog box is displayed. You cannot start a new import of files into the Teamcenter database until the previous one is completed or canceled.

Using the three options on the notification dialog box you can restart, suspend, or cancel the upload process.

Suspend

Stops processing and exits the Add To Teamcenter utility, but leaves the import queue intact. Data already created in Teamcenter is not deleted. When you later restart the Add To Teamcenter process using Retry, the files resume loading from this point.

Retry

Restarts a suspended transaction using the existing queue of documents. Pressing Return also restarts the process.

Cancel

Ends the transaction. Any documents that were imported remain in the Teamcenter database. The remaining documents are cleared from the queue.

If you previously suspended the transaction, the notification dialog box is displayed automatically the next time you run the Add To Teamcenter utility.

Direct and Indirect Documents

The documents that you select to add to the managed library are called direct documents. These direct documents may contain links to other documents or reference other documents, which are known as indirect documents. For example, when you add an assembly to a managed library, all the parts and subassemblies

within the selected assembly are indirect documents, if you select only the assembly document to add to the library. Similarly, a part copy within a part document or a part referenced within a drawing document are also indirect documents.

Add a document to Teamcenter

- Step 1:** On the Manage menu in Solid Edge, click Cache Assistant.
- Step 2:** In the Cache Assistant dialog box, select **Delete All** to clear your cache.
- Step 3:** Exit Solid Edge.
-  You should always exit Solid Edge prior to running Add to Teamcenter.
- Step 4:** In the \Solid Edge V20\Program folder, run *AddToTeamcenter.exe*.
- Step 5:** Connect to Teamcenter by entering the required information on the Login to Teamcenter dialog box.
- Step 6:** In the Add to Teamcenter dialog box, select the folder or document you want to add to the managed library.
- Step 7:** Click the **Add** button to add the selected document(s) to the Folders and Documents To Be Added list.
- Step 8:** To link the files that are added to Teamcenter to a specific folder, select the **Add Documents to this Folder** check box.
- You can use the **Browse** button to select a folder from the Solid Edge Embedded Client Select Folder dialog box.
-  Leaving the **Add Documents to This Folder** check box cleared will add the files to Teamcenter but they will not be linked to any folders in the database.
- Step 9:** (Optional) Click the **Dry Run** check box to examine the files to be added for broken links or missing information.
- The documents will not be added to the library and the documents will remain in the queue allowing you to make corrections and return to Add to Teamcenter.
- Upon returning to Add to Teamcenter you will be given the option to Restart the previous attempt using the existing queue of documents to be added, Suspend the transaction until a later time in which case the queue is maintained, or Discard the operation. Selecting Discard clears the queue.
- Step 10:** On the Add to Teamcenter dialog box, click **Add Documents to Library**.
- The Add to Teamcenter utility compares the Teamcenter import date with the date the files were last modified to see if the files have already been loaded into Teamcenter. If the date the files were modified is not newer than the import timestamp, it is assumed the file is already in

Teamcenter and it is disregarded. A message is included in the Add to Teamcenter log file.

You can load files over files that already exist in Teamcenter. Add to Teamcenter checks the item, revision, and dataset against those already in the database and gives you the option to check out the document and overwrite it, check out all files that exist and overwrite them, leave the file and use the file already in the database, leave all files and use the ones that already exist in the database or cancel the import process.

- Step 11:** (Optional) To check in the documents, on the Add to Teamcenter dialog box, select the **Update Status On All Documents to Checked-In** check box.
- Step 12:** (Optional) Reset the file properties value to the value selected in the **Set SEStatus To** portion of the dialog box.
- Step 13:** (Optional) Use a precise Bill of Material view when adding assemblies to Teamcenter by selecting the **Use precise BOM View when adding Assemblies** check box.
- Step 14:** In the Add to Teamcenter dialog box, click OK.

A log file is created in the *\Application Data\Unigraphics Solutions\Solid Edge\Version 20\Log Files* folder. You should work with your system administrator to establish a process for archiving or removing the log files on a periodic basis.

Add to Teamcenter dialog box

Specifies the folders and documents you want to add into a managed library for use with Teamcenter and Solid Edge Embedded Client (SEEC).

Look In

Displays the drive and folder in which you want to search for the document you want to add to the managed environment.

Go to Last Folder Visited

Returns to the last folder that you visited.

Up One Level

Moves the Look In folder up one level.

Views

Controls the display method for the listed documents.

Large Icons

Displays large icons for the documents.

Small Icons

Displays small icons for the documents.

List

Lists the names of the documents in columns.

Details

Displays a detailed view of the directory contents. The columns displayed include Name, Size, and Modified.

Name

Lists the folders and documents you can import into the library.

Preview Pane

Displays a preview image of the document if a preview bitmap was saved for the file.

To create a preview image, you can use the options on the Preview page of the File Properties dialog box. To learn how to do this, see the Help topic, *Create a Preview Image of a Document*.

Add

Adds the selected folders and documents to the Folders and Documents to be Added box.

Remove

Removes the selected folders and documents from the Folders and Documents to be Added box.

Remove All

Removes all folders and documents from the Folders and Documents to be Added box.

Include Subfolders

Imports the associated subfolders for the selected folders into the managed library.

Types

Sets the document types that will be added to the library from the selected folder(s).

- Draft documents (*.dft)
- Assembly documents (*.asm)
- Part documents (*.par)
- Sheet metal documents (*.psm)
- Weldment documents (*.pwd)
- All documents including Non-Solid Edge (*.*)

Folders and Documents to be Added

Lists the folders and documents you want to add to the managed environment.

Destination

Defines the location and status for the documents and folders you are adding to the library.

Add Documents to This Folder

Selecting the check box adds the files to Teamcenter and links them to folder specified.



You must have write access to the location specified

Selecting the check box with no folder specified will add the files to your default folder

Leaving the box unchecked adds the files to Teamcenter, but does not link the files to any folders in the database.

Browse for Library Folder

Displays the Select Folder dialog and allows the Teamcenter database to be browsed to find a folder.

New	Allows you to name and provide a description for a new folder.
No Folder	Dismisses the Select Folder dialog without a selected folder. Selecting No Folder does not uncheck the Add Documents to this Folder check box.
Find	Displays the Find dialog that allows you to search for a specific folder using various criteria.

Dry Run (Documents will not be added to the Library)

Checks the selected documents for illegal characters, corrupted files and broken links without adding the documents to the managed environment. Results of the dry run are written to a file for review.

Add Documents to the Library

Adds the selected documents to the library without checking for broken links.

Update Status on All Documents to be Checked-In

Automatically checks in the documents you are adding to the managed library.

Set SEStatus to:

Sets the file's Solid Edge status value to the value specified.



This value sets the status in Solid Edge. It is recommended that you set the status to Available unless there is a well thought-out strategy for document control.

Available

Any author can access the selected document. An author can change the status of the document to one of the following: In Work, In Review, Released, Baselined, or Obsolete. If the active document is not part of a set, you can change the status of the document from Available to any other state.

Baselined

The selected document is frozen from further edits. Once baselined, the document is read-only. Although you cannot change this document,

you can use the Save As command in Solid Edge or the Set Action to Revise command in Insight Connect : Revision Manager to create a new revision of the document. The new revision will have its own life cycle. If the document is baselined and resides in a managed folder, then all users, including Coordinators, can open the document as read-only.

In Review

The document is being reviewed. Any user can edit the document. The status can be changed from In Review to any other status.

In Work

The selected document is being worked on. Other users will be able to check out the document, but it will be opened read-only in Solid Edge.

Obsolete

The document is no longer the latest revision, but is still available for viewing. A Coordinator can make changes to the document but other users are not allowed to make changes.

Released

The selected document has been approved. Once released, the document is read-only for everyone except a Coordinator. A coordinator can open a released document in a managed library with read-write access. Although other users cannot change this document, they can create a revision of this document. The new document will have its own life-cycle.

Use Precise BOM View When Adding Assemblies

Sets the Bill of Material (BOM) as fixed links.



Precise should be used when loading legacy Insight files or filenames that include a revision number.